

Lecture 7

Impulse and momentum principle

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MC2103
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7 Impulse and momentum principle

Impulsive Motion

The thrust of a rocket acts over a specific time period to give the rocket linear momentum.



7 Impulse and momentum principle

Impulsive Motion

The impulse applied to the railcar by the wall brings its momentum to zero. Crash tests are often performed to help improve safety in different vehicles.

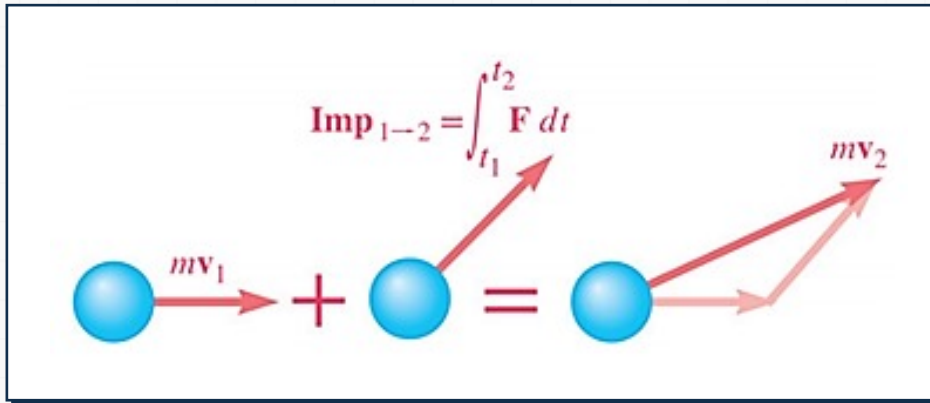


7 Impulse and momentum principle

Principle of Impulse and Momentum

⚙ Impulse of a force

- Dimensions of the impulse of a force are



Force $\times$ time

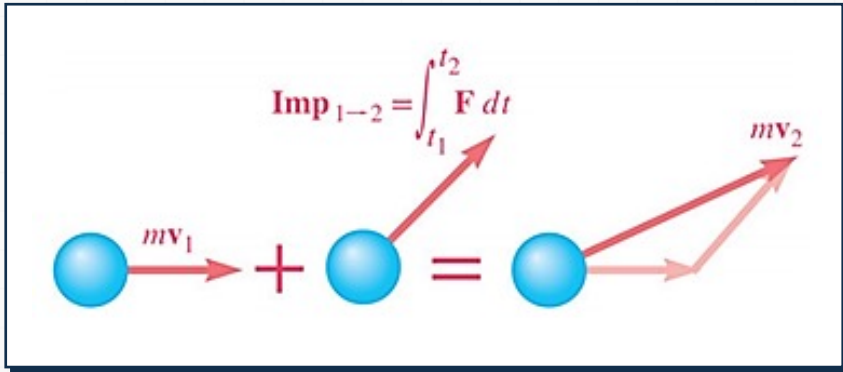
- Units for the impulse of a force are

$$\text{N} \cdot \text{s} = (\text{kg} \cdot \text{m}/\text{s}^2) \cdot \text{s} = \text{kg} \cdot \text{m}/\text{s}$$

7 Impulse and momentum principle

Principle of Impulse and Momentum

⚙ From Newton's second law



$$\vec{F} = \frac{d}{dt}(m\vec{v})$$

$m\vec{v}$ = linear momentum

$$\vec{F} dt = d(m\vec{v})$$

$$\int_{t_1}^{t_2} \vec{F} dt = m\vec{v}_2 - m\vec{v}_1$$

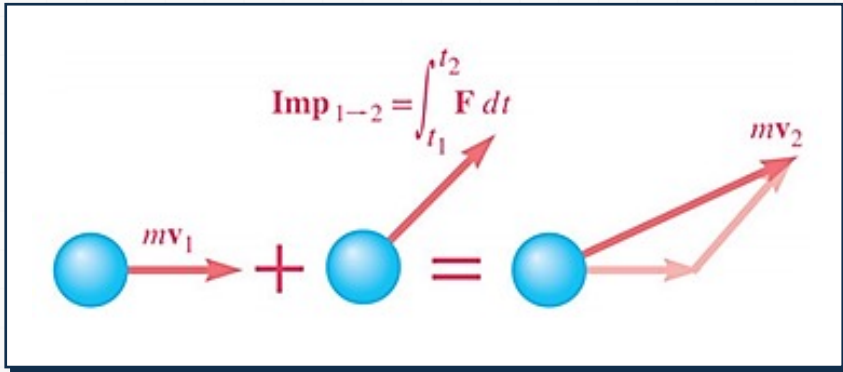
$$\int_{t_1}^{t_2} \vec{F} dt = \mathbf{Imp}_{1 \rightarrow 2} = \text{impulse of the force } \vec{F}$$

$$m\vec{v}_1 + \mathbf{Imp}_{1 \rightarrow 2} = m\vec{v}_2$$

7 Impulse and momentum principle

Principle of Impulse and Momentum

⚙ From Newton's second law

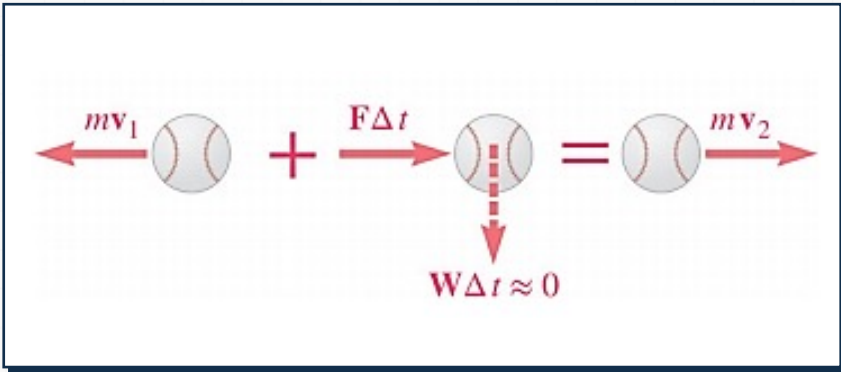


- The final momentum of the particle can be obtained by adding vectorially its initial momentum and the impulse of the force during the time interval.

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Impulsive Motion

⚙ Impulsive force

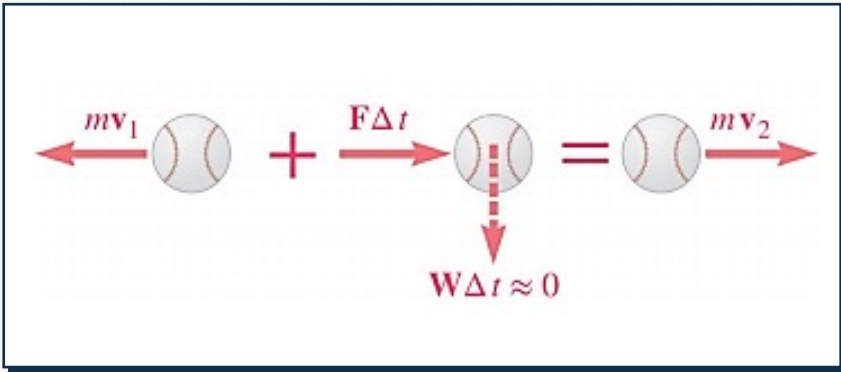


- Force acting on a particle during a very short time interval that is large enough to cause a significant change in momentum is called an impulsive force.

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Impulsive Motion

⚙ Impulsive force



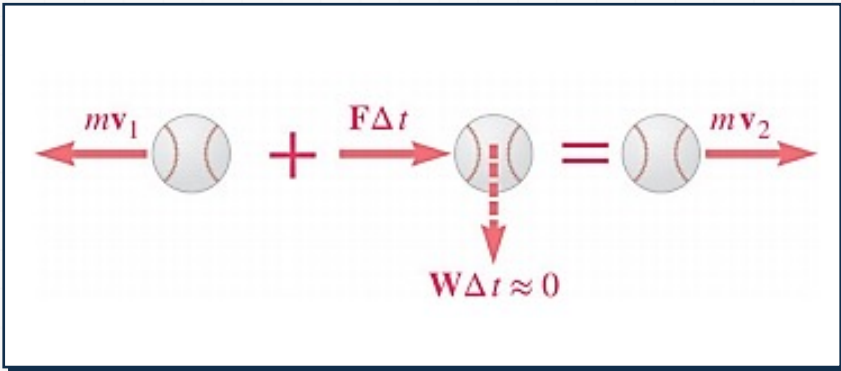
- When impulsive forces act on a particle,

$$m\vec{v}_1 + \sum \vec{F}\Delta t = m\vec{v}_2$$

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Impulsive Motion

⚙ Impulsive force



- When a baseball is struck by a bat
 - Contact occurs over a short time interval but force is large enough to change sense of ball motion.
- Nonimpulsive forces
 - **Nonimpulsive forces** are forces for which $\vec{F}\Delta t$ is small and therefore, may be neglected – an example of this is the weight of the baseball.

7 Impulse and momentum principle

Impact

The coefficient of restitution is used to characterize the “bounciness” of different sports equipment. The U.S. Golf Association limits the COR of golf balls at 0.83

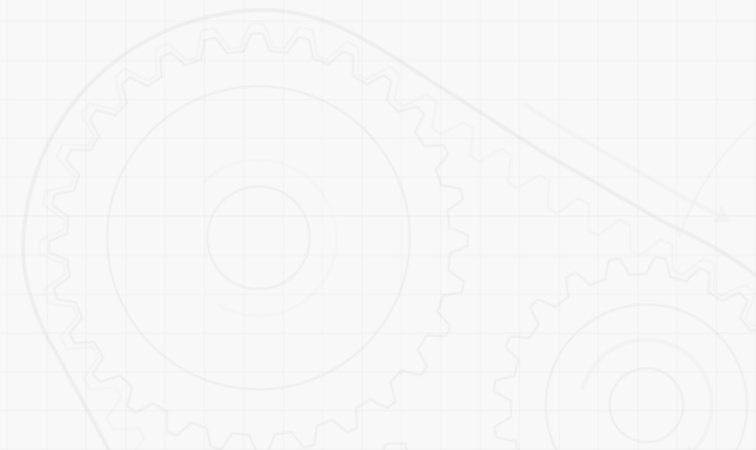


7 Impulse and momentum principle

Impact

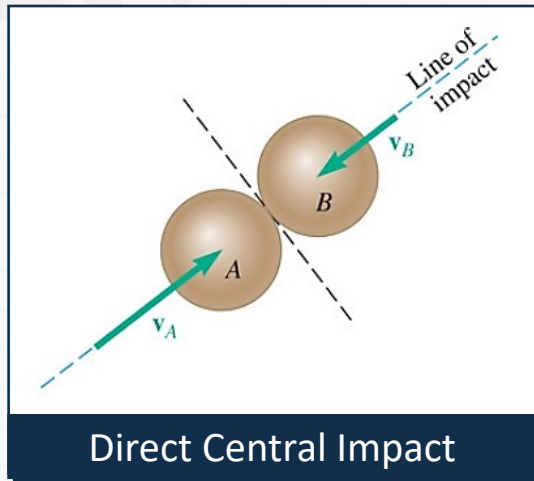
⚙ Definition of Impact

Collision between two bodies which occurs during a small time interval and during which the bodies exert large forces on each other.



7 Impulse and momentum principle

Impact



- Line of Impact

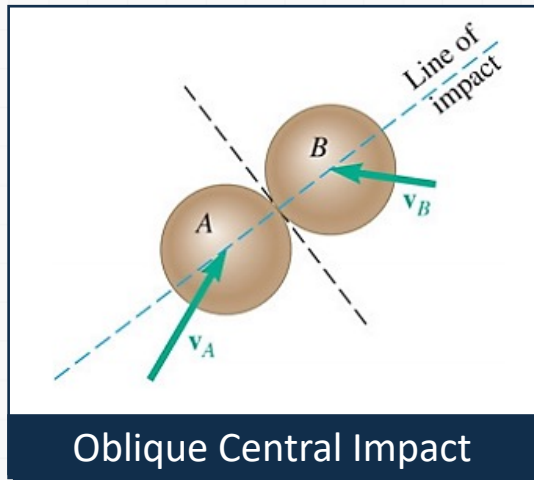
- Common normal to the surfaces in contact during impact.

- Central Impact

- Impact for which the mass centers of the two bodies lie on the line of impact; otherwise, it is an *eccentric impact*.

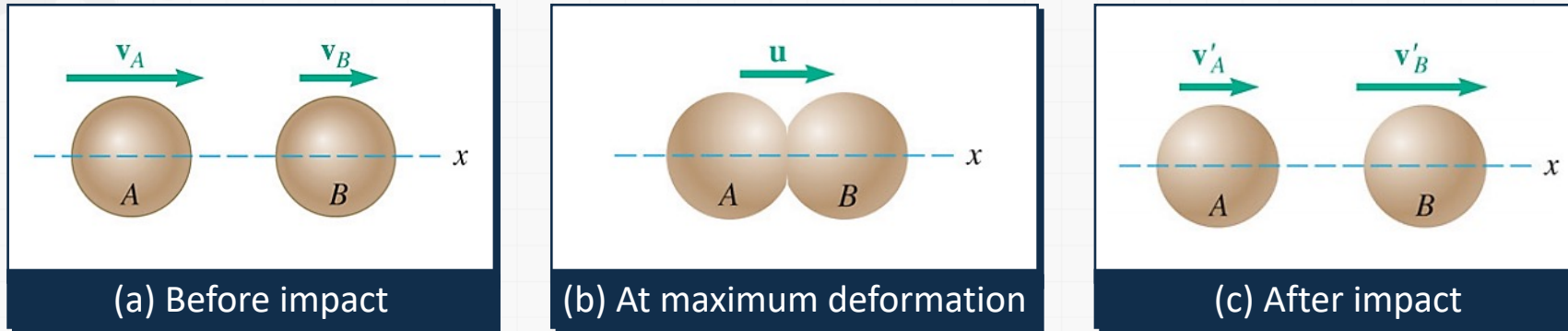
- Oblique Impact

- Impact for which one or both of the bodies move along a line other than the line of impact.



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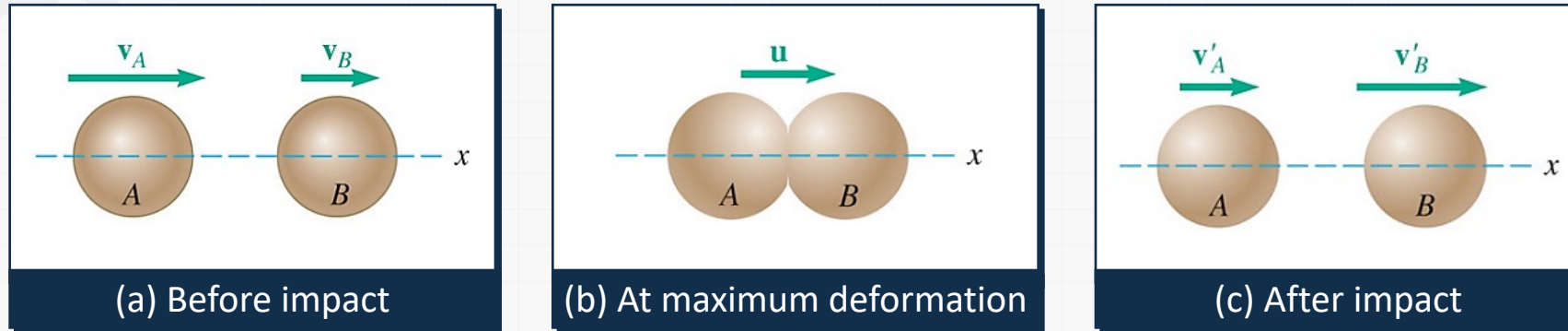
Direct Central Impact



- Bodies moving in the same straight line, $V_A > V_B$
- Upon impact the bodies
 - Undergo a **period of deformation**, at the end of which, they are in contact and moving at a common velocity.
- A **period of restitution**
 - Follows during which the bodies either regain their original shape or remain permanently deformed.

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Direct Central Impact



- Wish to determine

- The final velocities of the two bodies.

The total momentum of the two body system is preserved,

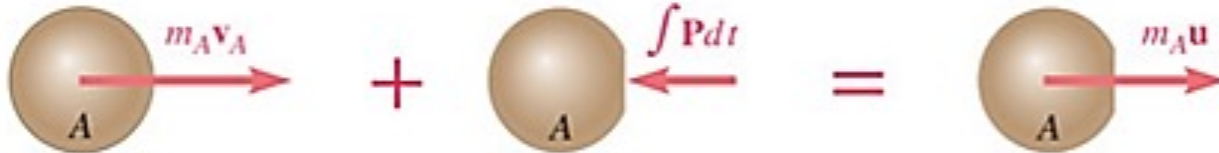
$$m_A v_A + m_B v_B = m_A v'_A + m_B v'_B$$

- A second relation between the final velocities is required.

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Direct Central Impact

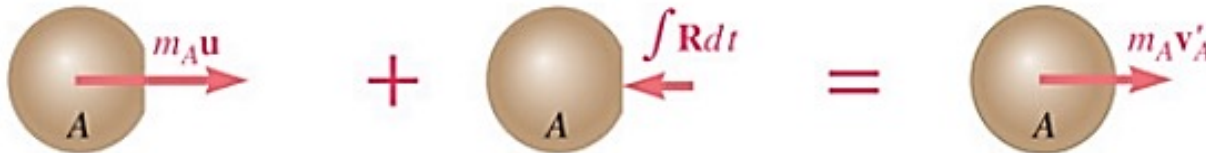
⚙ Period of deformation



(a) Period of deformation

$$m_A v_A - \int P dt = m_A u$$

⚙ Period of restitution



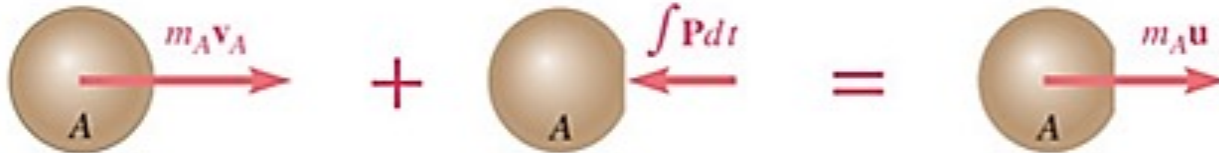
(b) Period of restitution

$$m_A u - \int R dt = m_A v'_A$$

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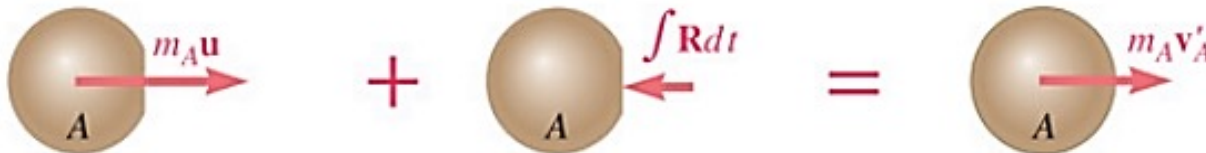
Direct Central Impact

⚙ Period of deformation



(a) Period of deformation

⚙ Period of restitution



(b) Period of restitution

e = coefficient of restitution

$$= \frac{\int R dt}{\int P dt} = \frac{u - v'_A}{v_A - u}$$

$$0 \leq e \leq 1$$

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Direct Central Impact

⚙️ A similar analysis of particle B yields

$$e = \frac{v'_B - u}{u - v_B}$$

⚙️ **Combining the relations**

- leads to the desired second relation between the final velocities.

$$v'_B - v'_A = e(v_A - v_B)$$

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Direct Central Impact

⚙️ **Perfectly plastic impact, $e = 0$:** $v'_B = v'_A = v'$

$$m_A v_A + m_B v_B = (m_A + m_B) v'$$

⚙️ **Perfectly elastic impact, $e = 1$:**

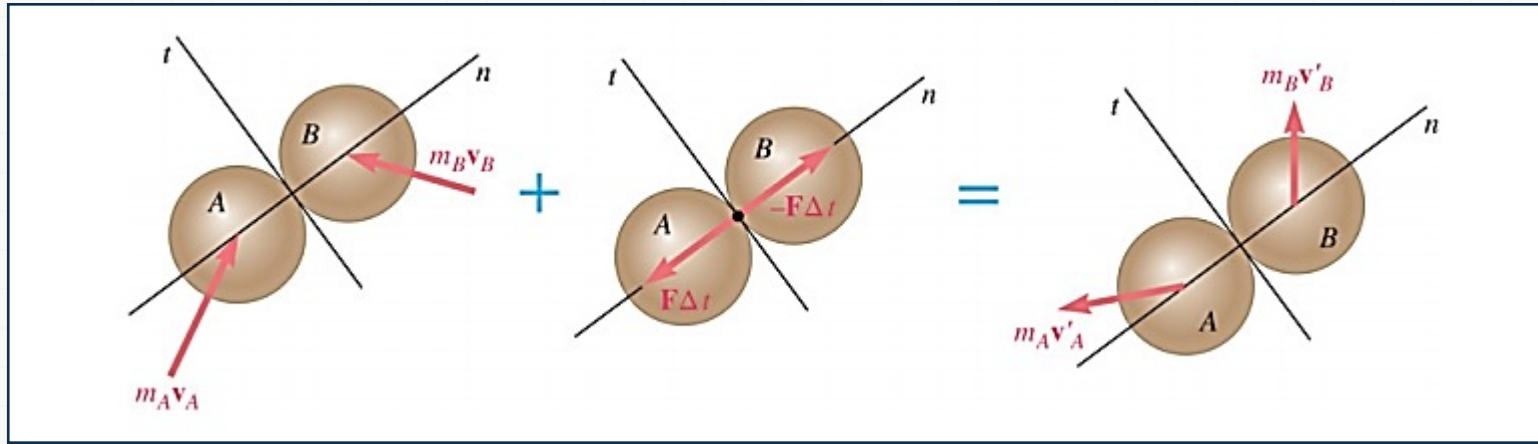
- Total energy and total momentum conserved.

$$v'_B - v'_A = v_A - v_B$$

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Oblique Central Impact

⚙️ Final velocities

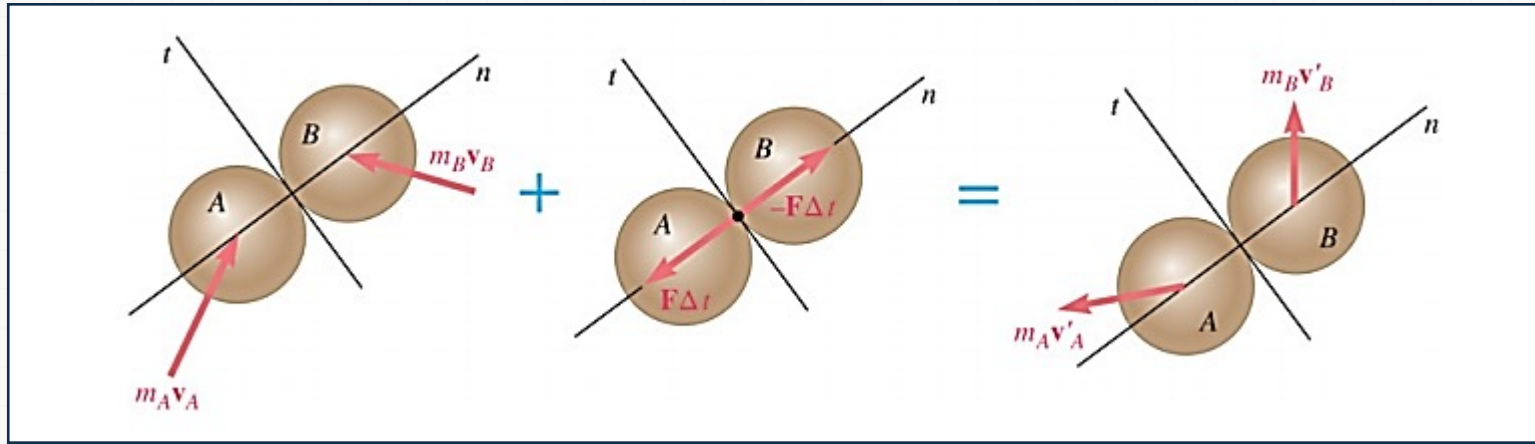


- Final velocities are unknown in magnitude and direction.
- Four equations are required.

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Oblique Central Impact

⚙️ No tangential impulse component



- Tangential component of momentum for each particle is conserved.

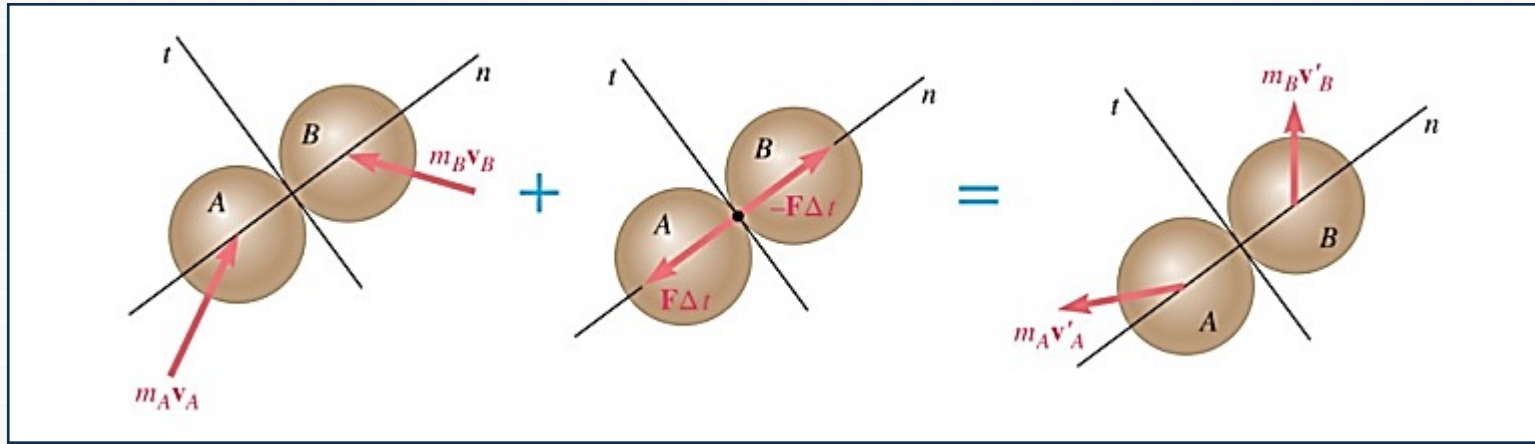
$$(v_A)_t = (v'_A)_t$$

$$(v_B)_t = (v'_B)_t$$

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Oblique Central Impact

⚙ Normal component of total momentum



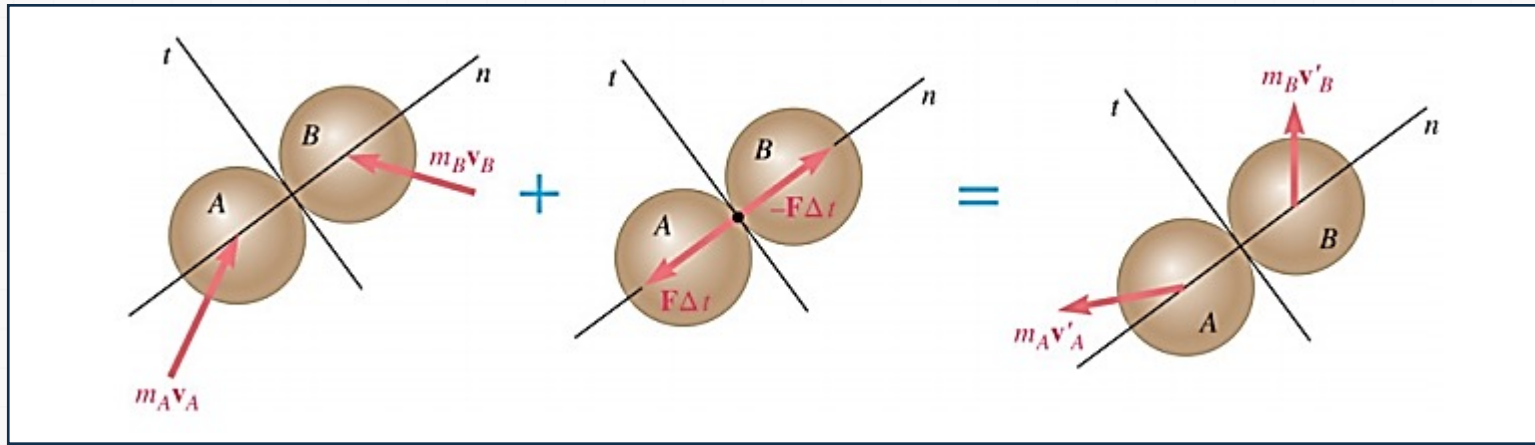
- Normal component of total momentum of the two particles is conserved.

$$m_A(v_A)_n + m_B(v_B)_n = m_A(v'_A)_n + m_B(v'_B)_n$$

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Oblique Central Impact

⚙ Normal components of relative velocities



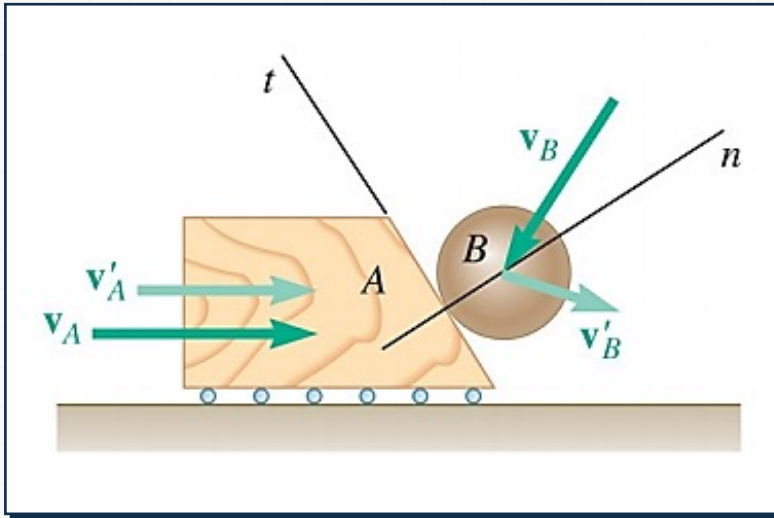
- Normal components of relative velocities before and after impact are related by the coefficient of restitution.

$$(v'_B)_n - (v'_A)_n = e((v_A)_n - (v_B)_n)$$

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Oblique Central Impact

⚙️ **Block constrained to move along horizontal surface.**

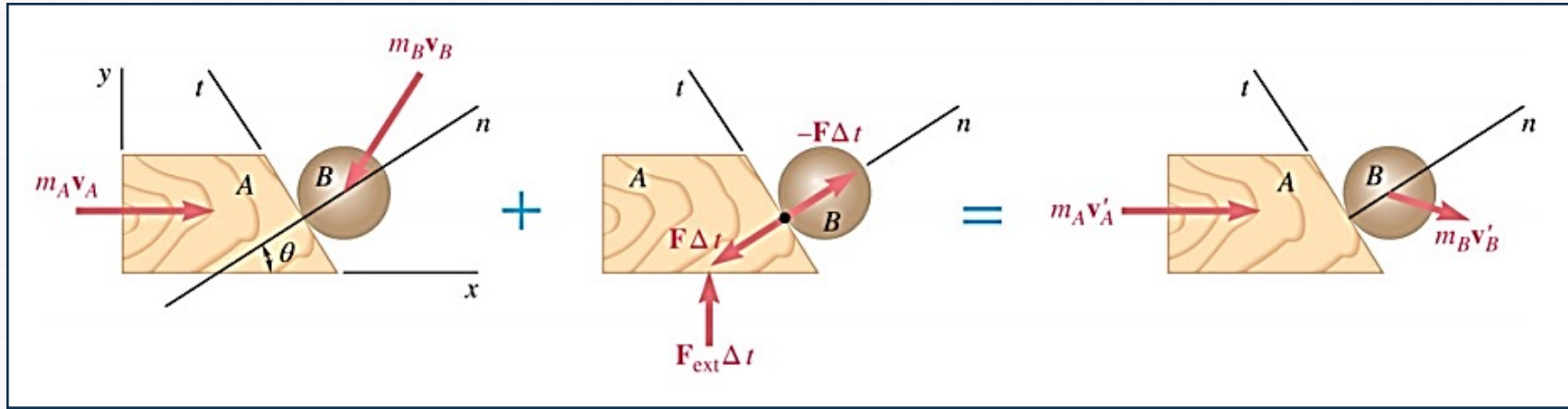


- Impulses from internal forces
 - $\vec{F}$ and $-\vec{F}$ along the n axis and from external force $\vec{F}_{ext}$ exerted by horizontal surface and directed along the vertical to the surface.
- Final velocity of ball
 - Final velocity of ball unknown in direction and magnitude and unknown final block velocity magnitude. Three equations required.

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Oblique Central Impact

⚙ Tangential momentum



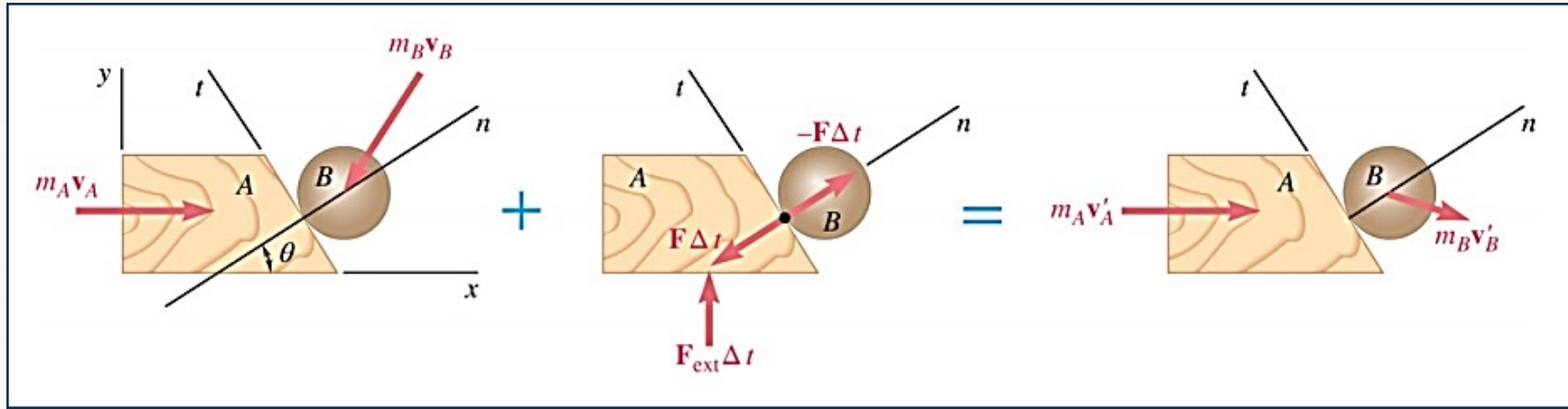
- Tangential momentum of ball is conserved.

$$(v_B)_t = (v'_B)_t$$

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Oblique Central Impact

⚙ Total horizontal momentum



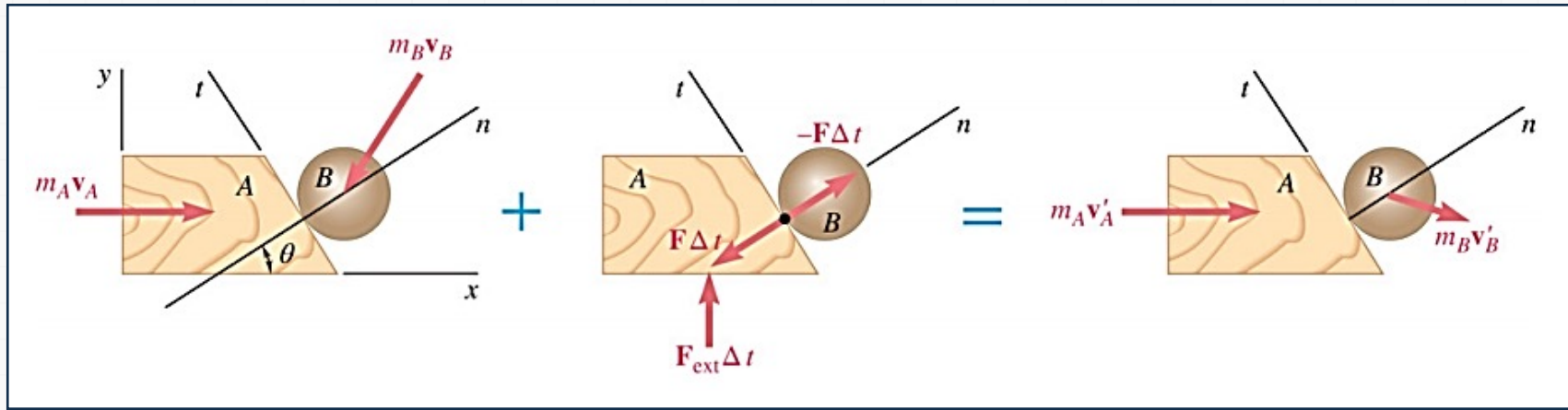
- Total horizontal momentum of block and ball is conserved.

$$m_A(v_A) + m_B(v_B)_x = m_A(v'_A) + m_B(v'_B)_x$$

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Oblique Central Impact

⚙ Normal component of relative velocities



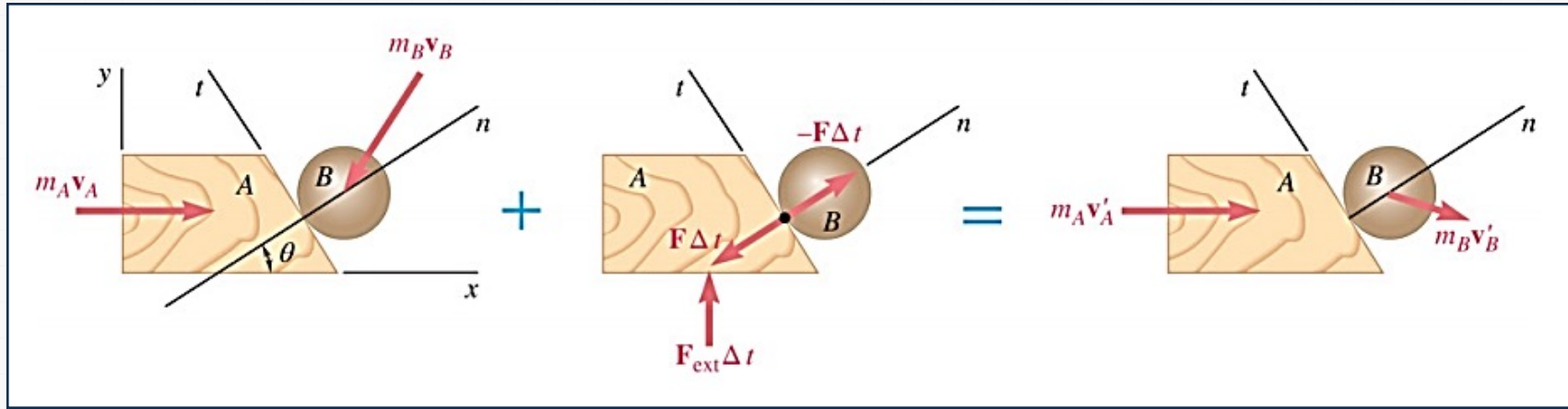
- Normal component of relative velocities of block and ball are related by coefficient of restitution.

$$(v'_B)_n - (v'_A)_n = e((v_A)_n - (v_B)_n)$$

7 Impulse and momentum principle

Oblique Central Impact

⚙ Note



- **Validity of last expression**

- Validity of last expression does not follow from previous relation for the coefficient of restitution.
- A similar but separate derivation is required.

7 Impulse and momentum principle

Problems Involving Multiple Principles

⚙️ **Three methods for the analysis of kinetics problems:**

Direct application of Newton's second law.

Method of work and energy.

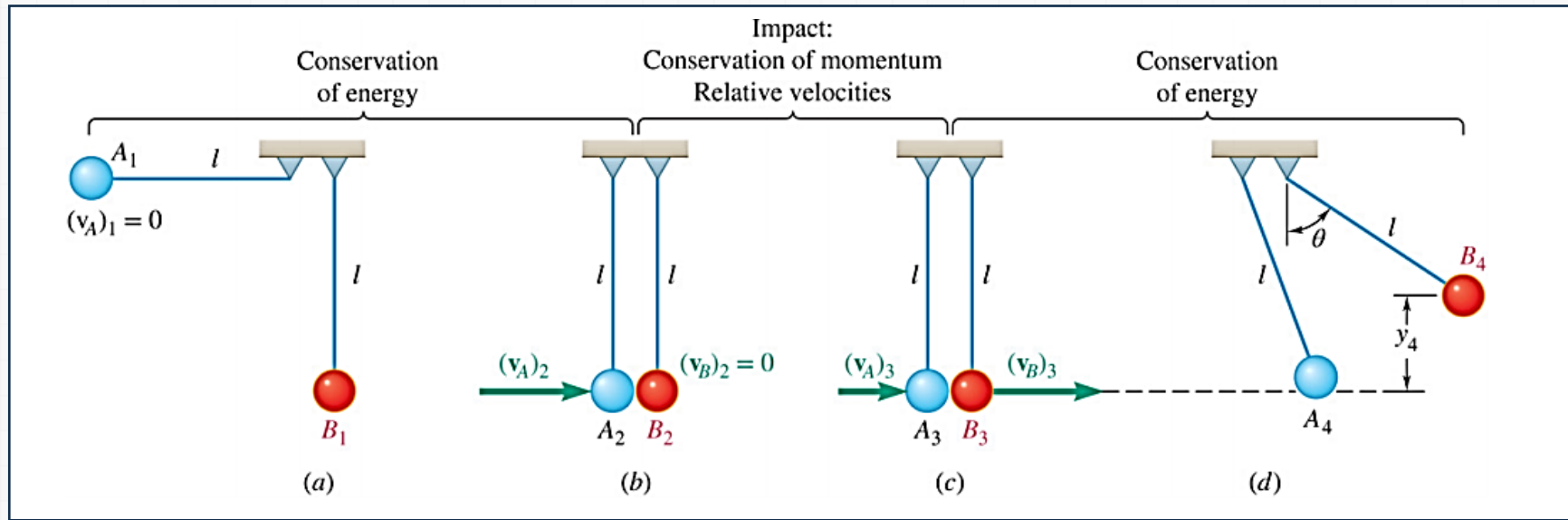
Method of impulse and momentum.

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Problems Involving Multiple Principles

⚙️ Select the method

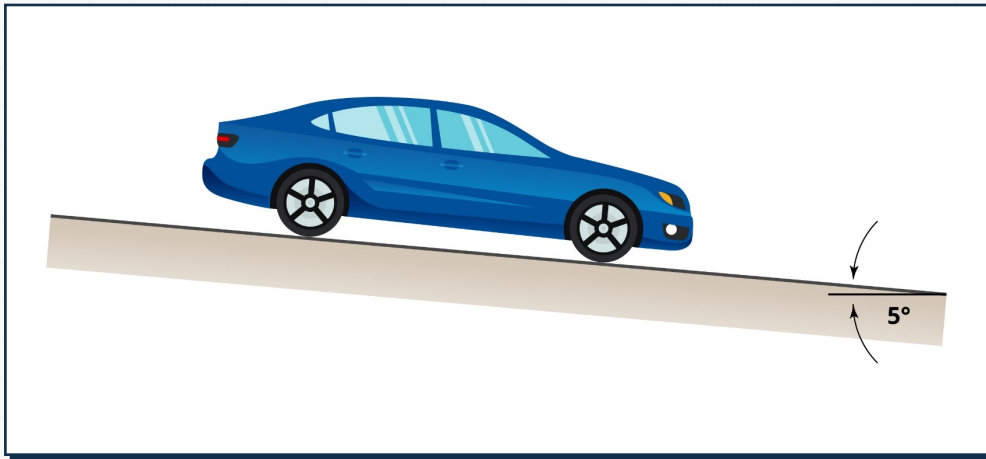
- Best suited for the problem or part of a problem under consideration.



7 Impulse and momentum principle

Sample Problem 1

⚙ Determine the time



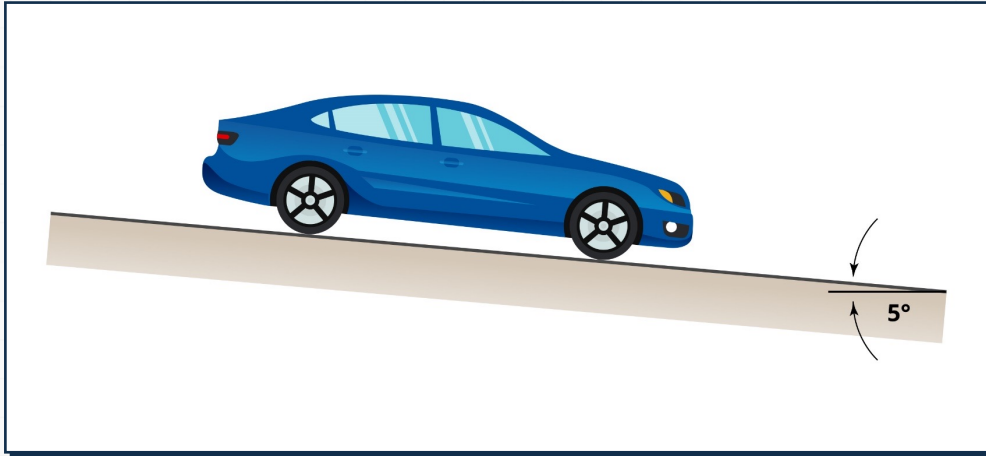
- An automobile weighing 1800 kg is driven down a 5° incline at a speed of 100 km/h when the brakes are applied, causing a constant total braking force of 7000 N .
- Determine the time required for the automobile to come to a stop.

7 Impulse and momentum principle

Sample Problem 1

⚙ Determine the time

• Strategy



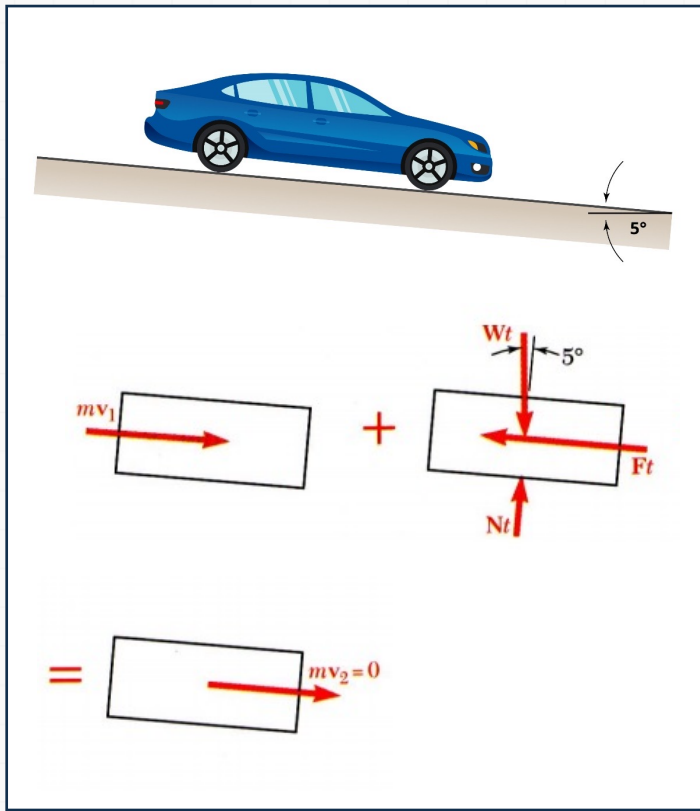
- Apply the principle of impulse and momentum. The impulse is equal to the product of the constant forces and the time interval.

7 Impulse and momentum principle

Sample Problem 1

⚙ Determine the time

• Modeling and Analysis



- Apply the principle of impulse and momentum.

$$m\vec{v}_1 + \sum \text{Imp}_{1 \rightarrow 2} = m\vec{v}_2$$

- Taking components parallel to the incline,

$$mv_1 + (mg \sin 5^\circ)t - Ft = 0$$

$$100 \frac{\text{km}}{\text{h}} = 100 \frac{\text{km}}{\text{h}} \left(\frac{1000\text{m}}{1\text{km}} \right) \left(\frac{1\text{h}}{3600\text{s}} \right) = 27.78\text{m/s}$$

$$(1800)(27.78) + (1800)(9.81) \sin 5^\circ t - (7000)t = 0$$

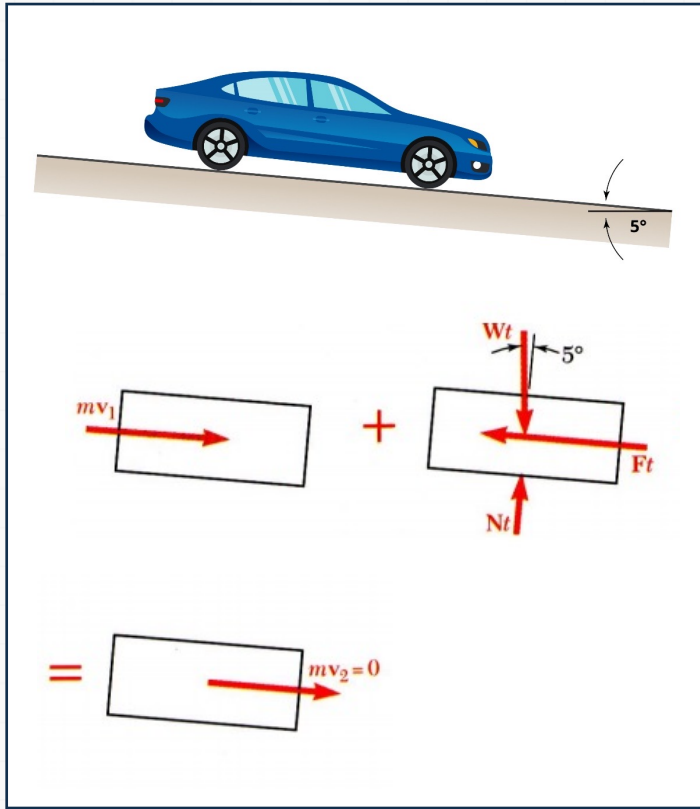
$$t = 9.16 \text{ s}$$

7 Impulse and momentum principle

Sample Problem 1

⚙ Determine the time

• Reflect and Think

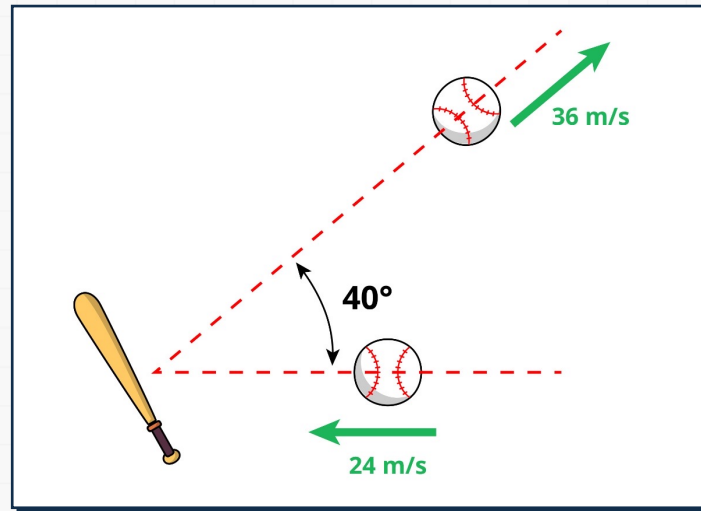


- You could use Newton's second law to solve this problem. First, you would determine the car's deceleration, separate variables, and then integrate $a = dv/dt$ to relate the velocity, deceleration, and time.
- You could not use conservation of energy to solve this problem, because this principle does not involve time.

7 Impulse and momentum principle

Sample Problem 2

⚙ Determine the average impulsive force



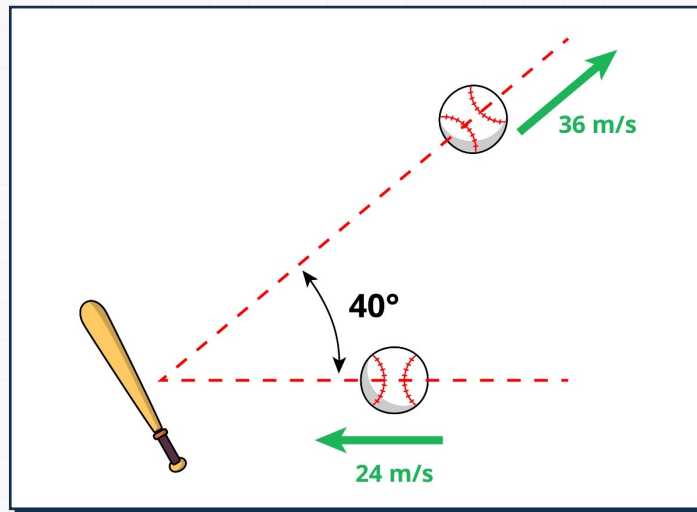
- A 120 g baseball is pitched with a velocity of 24 m/s . After the ball is hit by the bat, it has a velocity of 36 m/s in the direction shown.
- If the bat and ball are in contact for 0.015 s , determine the average impulsive force exerted on the ball during the impact.

7 Impulse and momentum principle

Sample Problem 2

⚙ Determine the time

• Strategy



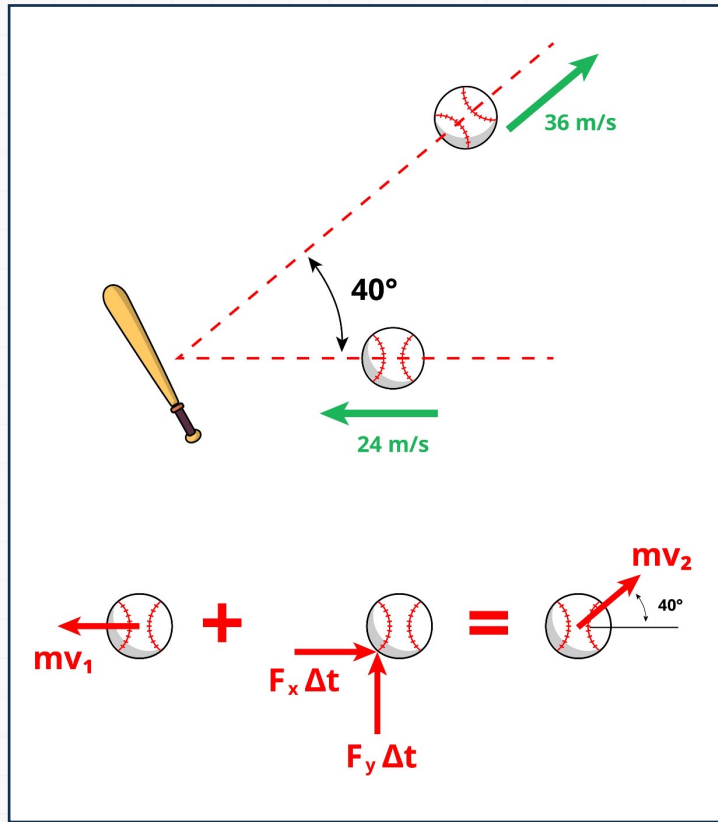
- Apply the principle of impulse and momentum in terms of horizontal and vertical component equations.

7 Impulse and momentum principle

Sample Problem 2

⚙ Determine the time

• Modeling and Analysis



- Apply the principle of impulse and momentum in terms of horizontal and vertical component equations.

$$m\vec{v}_1 + \sum \mathbf{Imp}_{1 \rightarrow 2} = m\vec{v}_2$$

x component equation:

$$-mv_1 + F_x \Delta t = mv_2 \cos 40^\circ$$

$$-(0.12)(24) + F_x(0.015) = (0.12)(36 \cos 40^\circ)$$

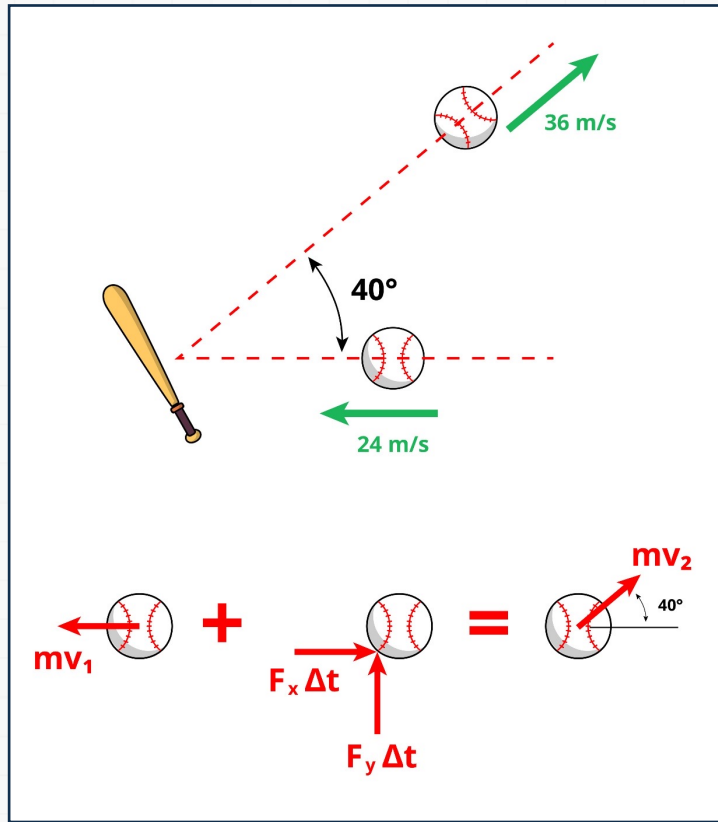
$$F_x = +412.6 \text{ N}$$

7 Impulse and momentum principle

Sample Problem 2

⚙ Determine the time

• Modeling and Analysis



- y component equation:

$$0 + F_y \Delta t = mv_2 \sin 40^\circ$$

$$F_y(0.015) = (0.12)(36 \sin 40^\circ)$$

$$F_y = +185.1 \text{ N}$$

$$\vec{F} = (413\text{N})\hat{i} + (185.1\text{N})\hat{j}$$

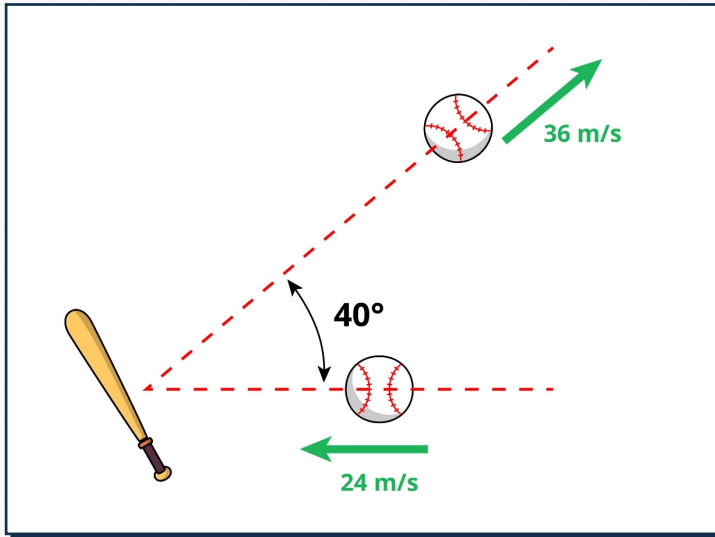
$$F = 452\text{N}$$

7 Impulse and momentum principle

Sample Problem 2

⚙ Determine the time

• Reflect and Think

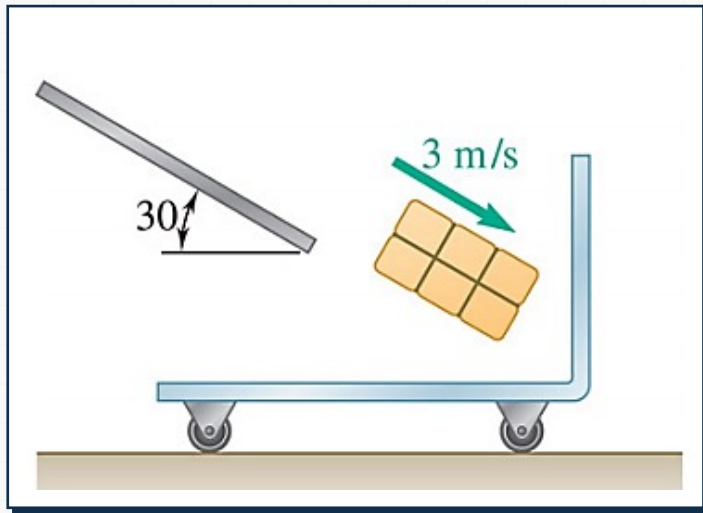


- In this problem, we neglected the impulse due to the weight. This would have had a magnitude of $(0.12\text{kg}) \left(9.81 \frac{\text{m}}{\text{s}^2}\right) (0.015\text{s}) = 0.01766 \text{ N} \cdot \text{s}$.
- This indeed is much smaller than the impulse exerted on the ball by the bat, which is $(452 \text{ N})(0.015 \text{ s}) = 6.78 \text{ N} \cdot \text{s}$.

7 Impulse and momentum principle

Sample Problem 3

⚙ Determine the velocity, impulse, and fraction



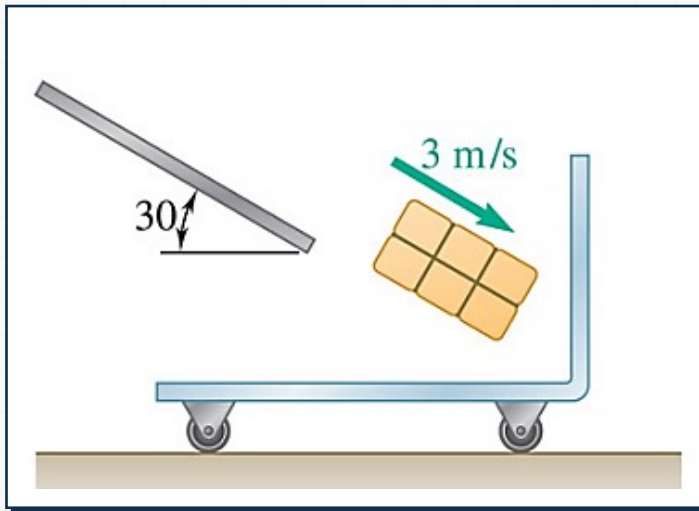
- A 10 kg package drops from a chute into a 24 kg cart with a velocity of 3 m/s .
- Knowing that the cart is initially at rest and can roll freely, determine
 - (a) the final velocity of the cart,
 - (b) the impulse exerted by the cart on the package, and
 - (c) the fraction of the initial energy lost in the impact.

7 Impulse and momentum principle

Sample Problem 3

⚙ Determine the velocity, impulse, and fraction

• Strategy



- Apply the principle of impulse and momentum to the package-cart system to determine the final velocity.
- Apply the same principle to the package alone to determine the impulse exerted on it from the change in its momentum.

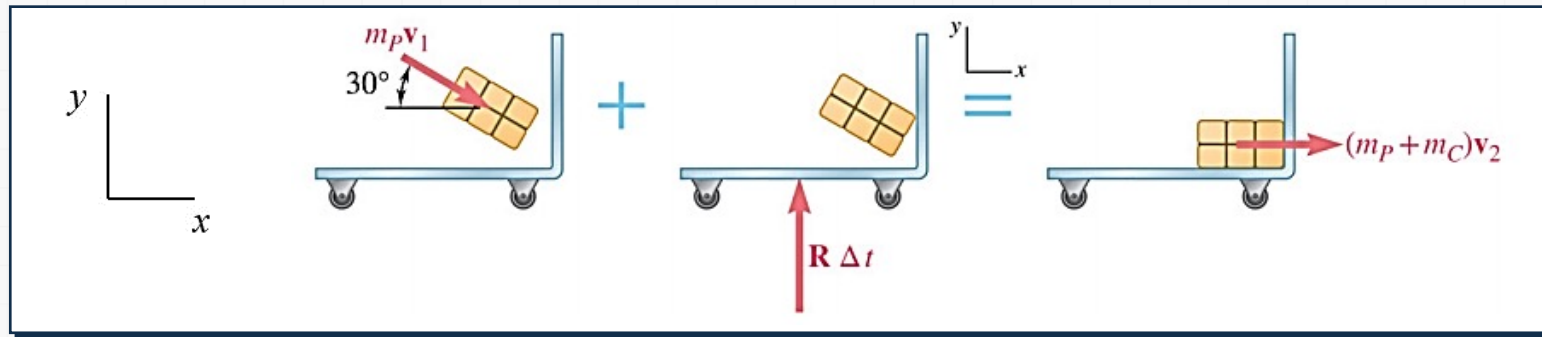
7 Impulse and momentum principle

Sample Problem 3

⚙ Determine the velocity, impulse, and fraction

• Modeling and Analysis

- Apply the principle of impulse and momentum to the package-cart system to determine the final velocity.



$$m_p \vec{v}_1 + \sum \mathbf{Imp}_{1 \rightarrow 2} = (m_p + m_c) \vec{v}_2$$

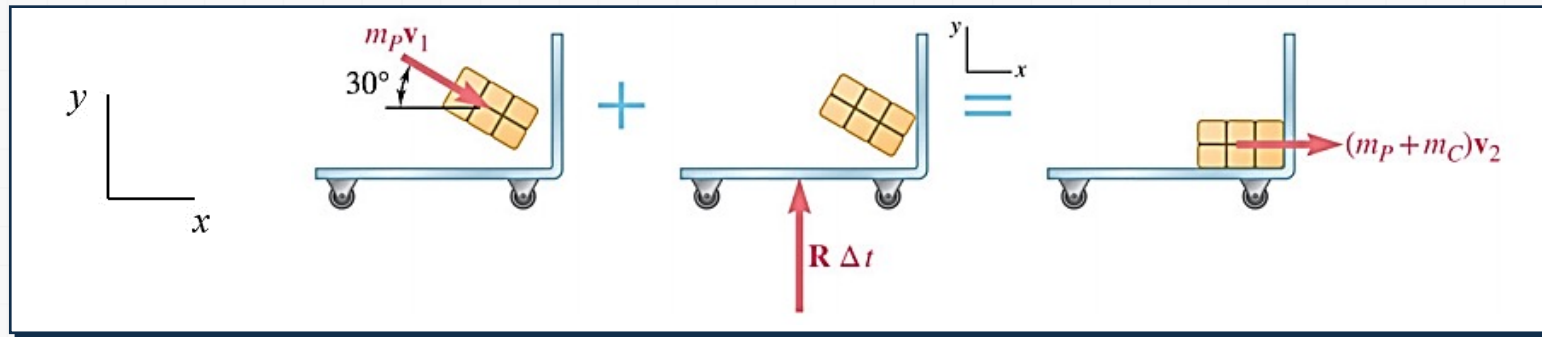
7 Impulse and momentum principle

Sample Problem 3

⚙ Determine the velocity, impulse, and fraction

• Modeling and Analysis

- Apply the principle of impulse and momentum to the package-cart system to determine the final velocity.



x components:

$$m_p v_1 \cos 30^\circ + 0 = (m_p + m_c) v_2$$

$$(10)(3)\cos 30^\circ = (10 + 25)v_2$$

$$v_2 = 0.742 \text{ m/s}$$

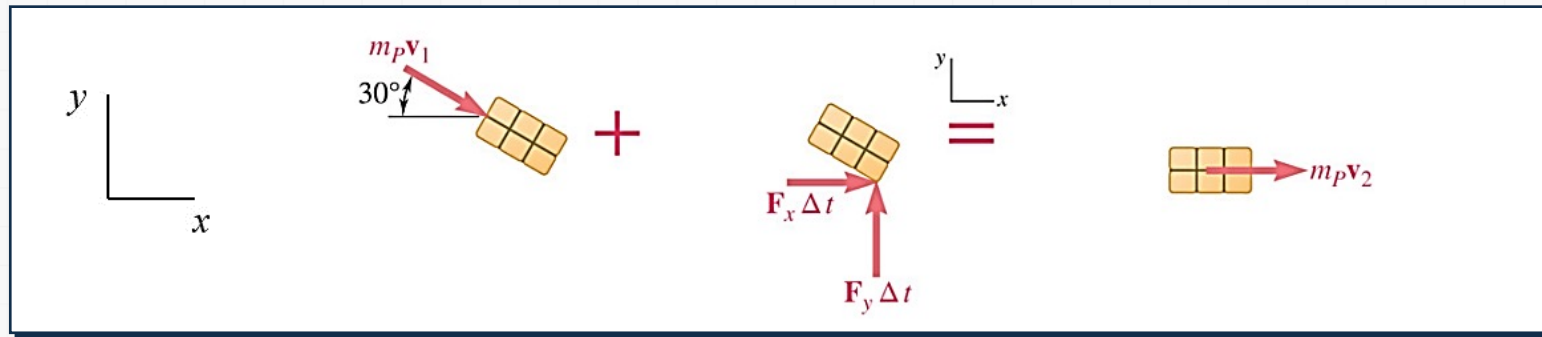
7 Impulse and momentum principle

Sample Problem 3

⚙ Determine the velocity, impulse, and fraction

• Modeling and Analysis

- Apply the same principle to the package alone to determine the impulse exerted on it from the change in its momentum.



$$m_p \vec{v}_1 + \sum \text{Imp}_{1 \rightarrow 2} = m_p \vec{v}_2$$

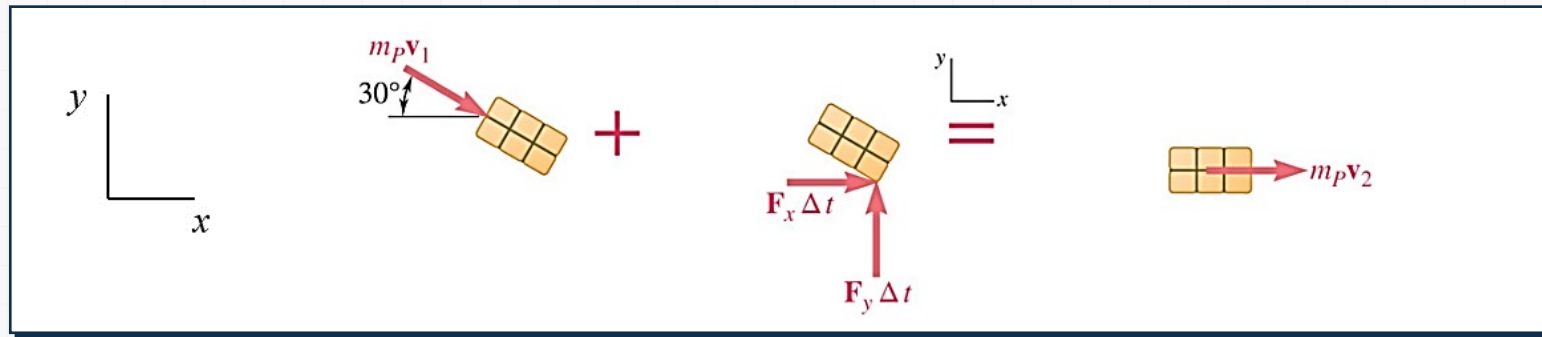
7 Impulse and momentum principle

Sample Problem 3

⚙ Determine the velocity, impulse, and fraction

• Modeling and Analysis

- Apply the same principle to the package alone to determine the impulse exerted on it from the change in its momentum.



x components:

$$m_p v_1 \cos 30^\circ + F_x \Delta t = m_p v_2$$

$$(10)(3) \cos 30^\circ = F_x \Delta t = (10) v_2$$

$$F_x \Delta t = -18.56 \text{ N} \cdot \text{s}$$

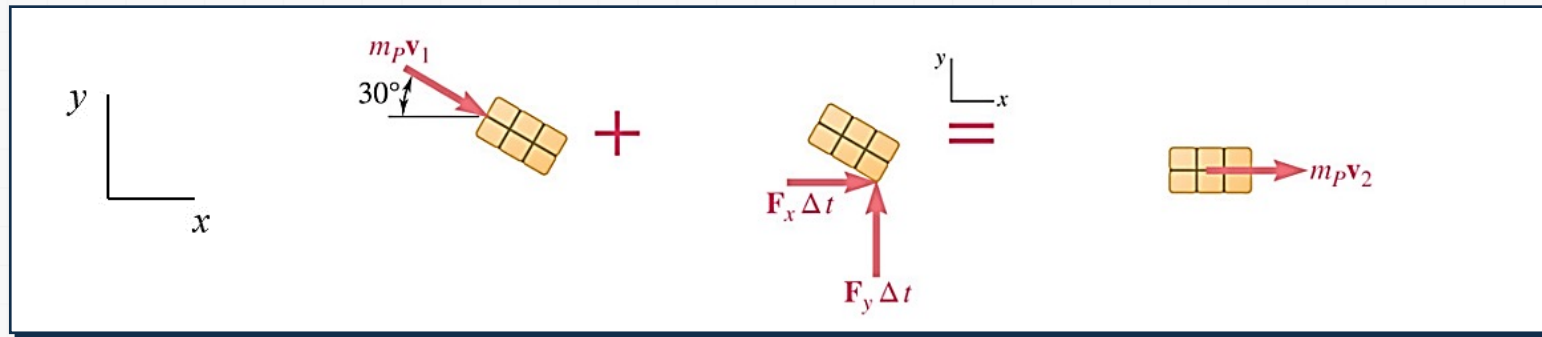
7 Impulse and momentum principle

Sample Problem 3

⚙ Determine the velocity, impulse, and fraction

• Modeling and Analysis

- Apply the same principle to the package alone to determine the impulse exerted on it from the change in its momentum.



y components:

$$-m_p v_1 \sin 30^\circ + F_y \Delta t = 0$$

$$-(10)(3) \sin 30^\circ + F_y \Delta t = 0$$

$$F_y \Delta t = 15 \text{ N} \cdot \text{s}$$

$$\Sigma \text{Imp}_{1 \rightarrow 2} = \vec{F} \Delta t = (-18.56) \hat{i} + (15) \hat{j}$$

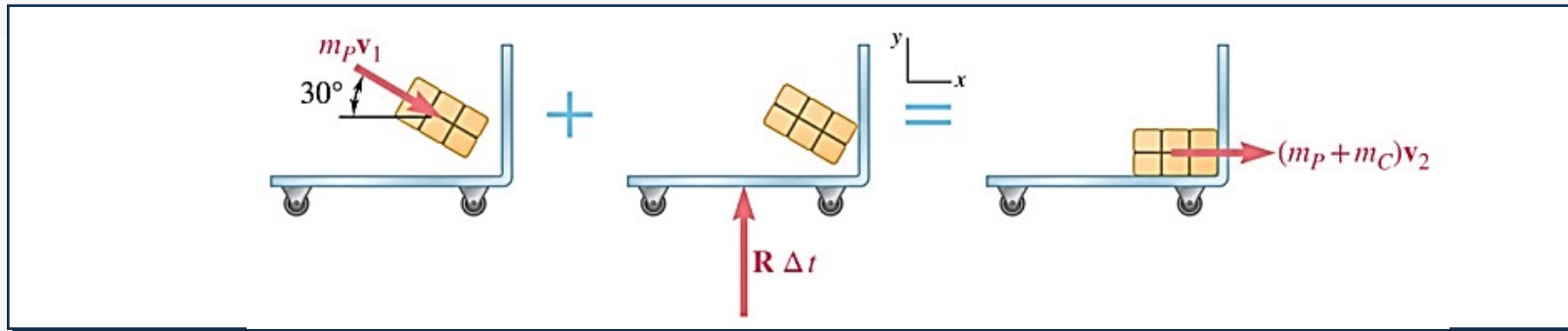
$$F \Delta t = 23.9 \text{ N} \cdot \text{s}$$

7 Impulse and momentum principle

Sample Problem 3

⚙️ Determine the velocity, impulse, and fraction

• Modeling and Analysis



To determine the fraction of energy lost,

$$T_1 = \frac{1}{2}m_p v_1^2 = \frac{1}{2}(10)(3)^2 = 45 \text{ J}$$

$$T_2 = \frac{1}{2}(m_p + m_c)v_2^2 = \frac{1}{2}(10 + 25)(0.742)^2 = 9.63 \text{ J}$$

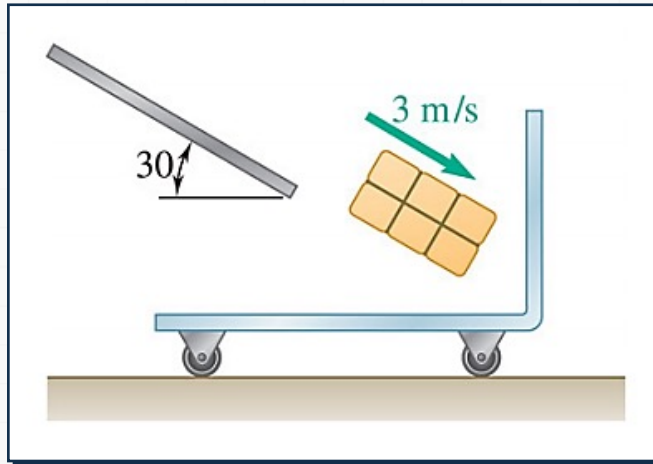
$$\frac{T_1 - T_2}{T_1} = \frac{45 - 9.63}{45} = 0.786$$

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Sample Problem 3

⚙️ **Determine the velocity, impulse, and fraction**

• Reflection and Think



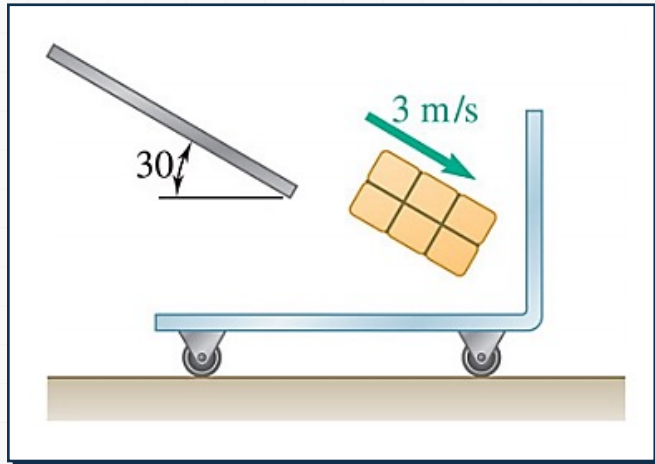
- Except in the purely theoretical case of a “perfectly elastic” collision, mechanical energy is never conserved in a collision between two objects, even though linear momentum may be conserved.

7 Impulse and momentum principle

Sample Problem 3

⚙️ **Determine the velocity, impulse, and fraction**

• Reflection and Think

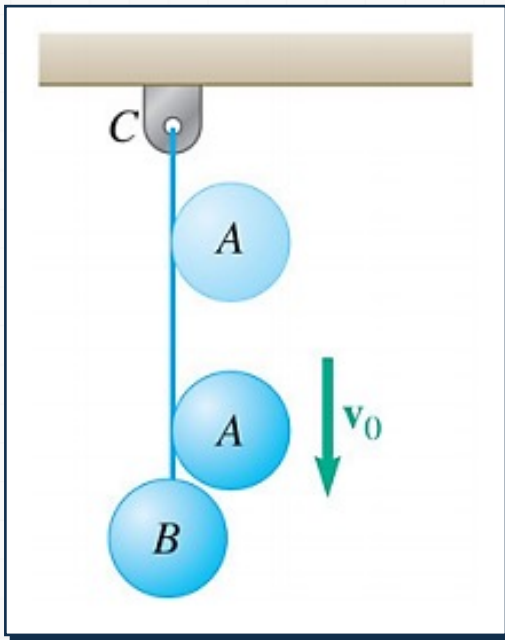


- Note that, in this problem, momentum was conserved in the x direction but was not conserved in the y direction because of the vertical impulse on the wheels of the cart. Whenever you deal with an impact, you need to use impulse-momentum methods.

7 Impulse and momentum principle

Sample Problem 4

⚙ Determine the velocity of each ball



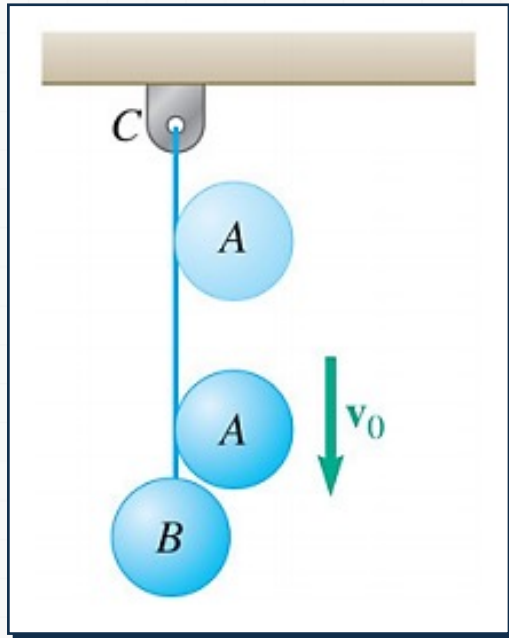
- Ball B is hanging from an inextensible cord. An identical ball A is released from rest when it is just touching the cord and acquires a velocity v_0 before striking ball B .
- Assuming perfectly elastic impact ($e = 1$) and no friction, determine the velocity of each ball immediately after impact.

7 Impulse and momentum principle

Sample Problem 4

⚙ Determine the velocity of each ball

• Strategy



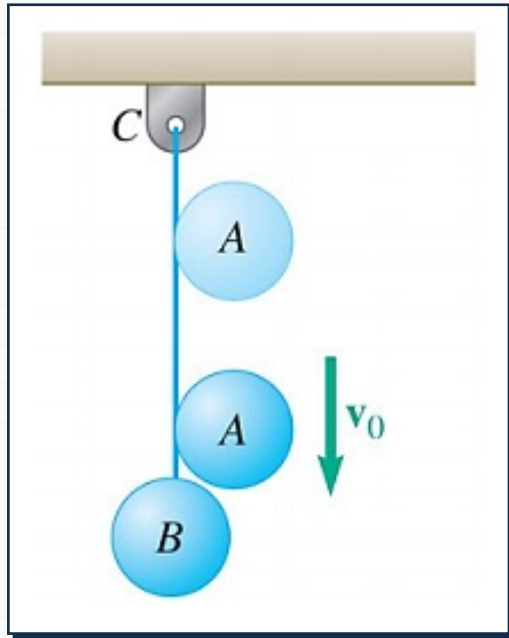
- Determine orientation of impact line of action.
- The momentum component of ball A tangential to the contact plane is conserved.
- The total horizontal momentum of the two ball system is conserved.

7 Impulse and momentum principle

Sample Problem 4

⚙ Determine the velocity of each ball

• Strategy



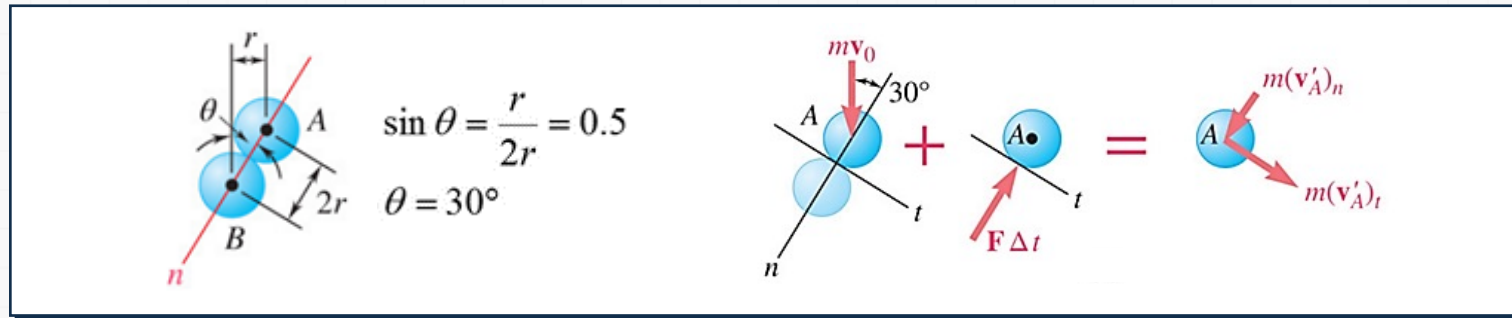
- The relative velocities along the line of action before and after the impact are related by the coefficient of restitution.
- Solve the last two expressions for the velocity of ball A along the line of action and the velocity of ball B which is horizontal.

7 Impulse and momentum principle

Sample Problem 4

⚙️ Determine the velocity of each ball

• Modeling and Analysis



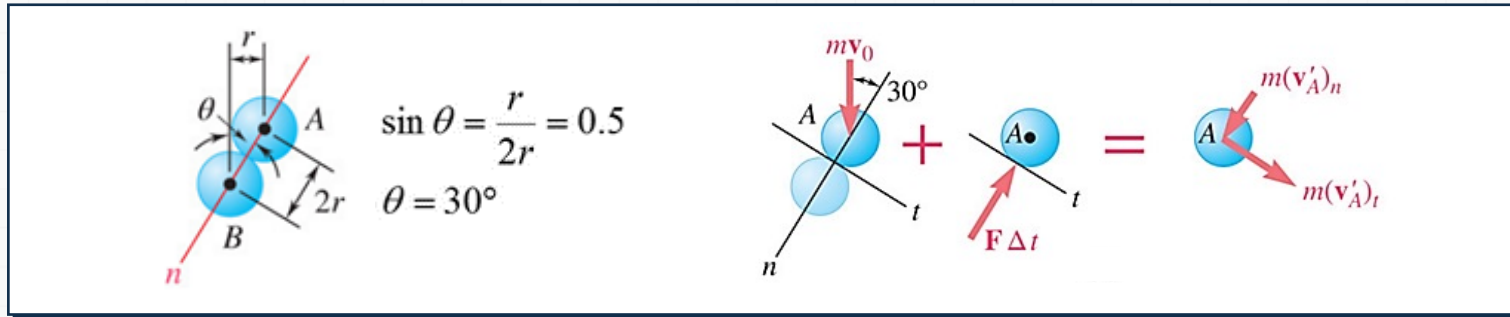
- Determine orientation of impact line of action.

7 Impulse and momentum principle

Sample Problem 4

⚙ Determine the velocity of each ball

• Modeling and Analysis



- The momentum component of ball A tangential to the contact plane is conserved.

$$m\vec{v}_A + \vec{F}\Delta t = m\vec{v}'_A$$

$$mv_0 \sin 30^\circ + 0 = m(v'_A)_t$$

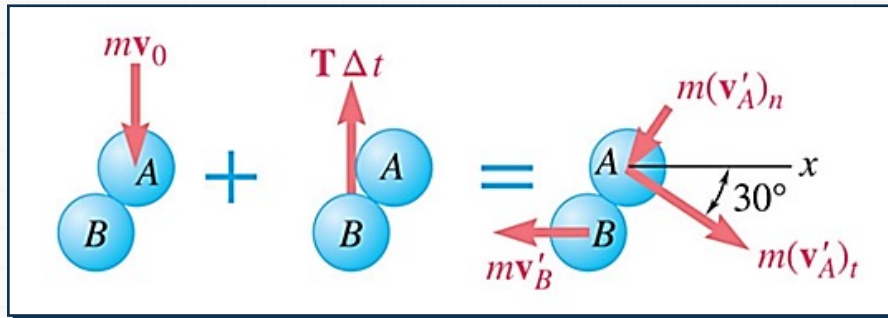
$$(v'_A)_t = 0.5 v_0$$

7 Impulse and momentum principle

Sample Problem 4

⚙ Determine the velocity of each ball

• Modeling and Analysis



- The total horizontal (x component) momentum of the two ball system is conserved.

$$m\vec{v}_A + \vec{T}\Delta t = m\vec{v}'_A + m\vec{v}'_B$$

$$0 = m(v'_A)_t \cos 30^\circ - m(v'_A)_n \sin 30^\circ - mv'_B$$

$$0 = (0.5v_0)\cos 30^\circ - (v'_A)_n \sin 30^\circ - v'_B$$

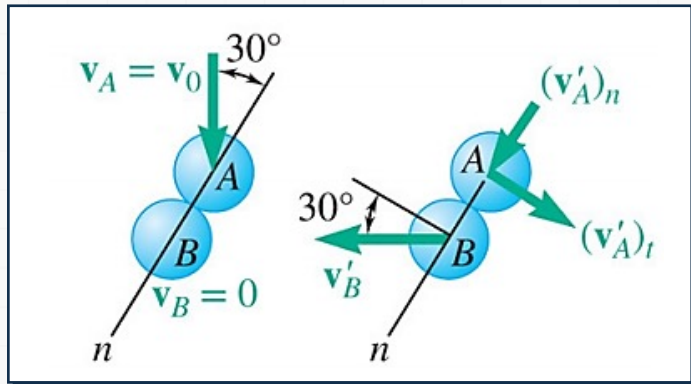
$$0.5(v'_A)_n + v'_B = 0.433v_0$$

7 Impulse and momentum principle

Sample Problem 4

⚙ Determine the velocity of each ball

• Modeling and Analysis



- The relative velocities along the line of action before and after the impact are related by the coefficient of restitution.

$$(v'_B)_n - (v'_A)_n = e((v_A)_n - (v_B)_n)$$

$$v'_B \sin 30^\circ - (v'_A)_n = v_0 \cos 30^\circ - 0$$

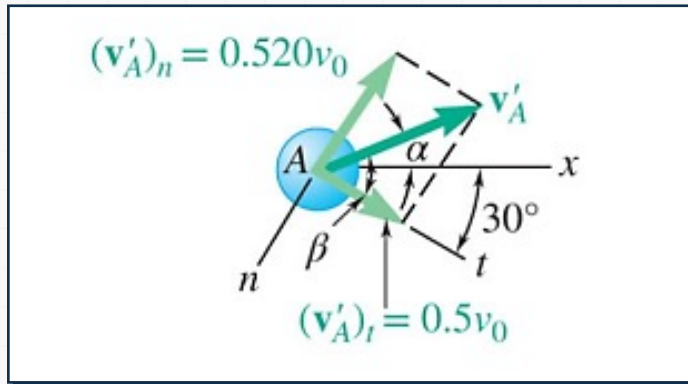
$$0.5v'_B - (v'_A)_n = 0.866v_0$$

7 Impulse and momentum principle

Sample Problem 4

⚙ Determine the velocity of each ball

• Modeling and Analysis



- Solve the last two expressions for the velocity of ball *A* along the line of action and the velocity of ball *B* which is horizontal.

$$(\vec{v}'_A)_n = -0.520v_0$$

$$v'_B = 0.693v_0$$

$$\vec{v}'_A = 0.5v_0\vec{\lambda}_t - 0.520v_0\vec{\lambda}_n$$

$$v'_A = 0.721v_0$$

$$\beta = \tan^{-1}\left(\frac{0.52}{0.5}\right) = 46.1^\circ$$

$$\alpha = 46.1^\circ - 30^\circ = 16.1^\circ$$

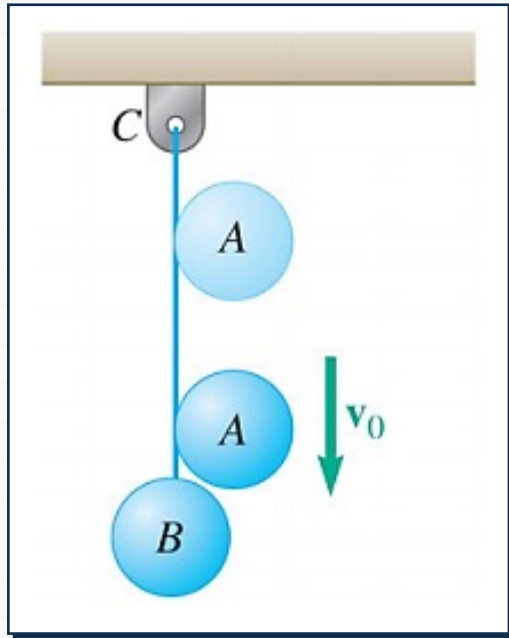
$$v'_B = 0.693v_0 \leftarrow$$

7 Impulse and momentum principle

Sample Problem 4

⚙ Determine the velocity of each ball

• Reflect and Think

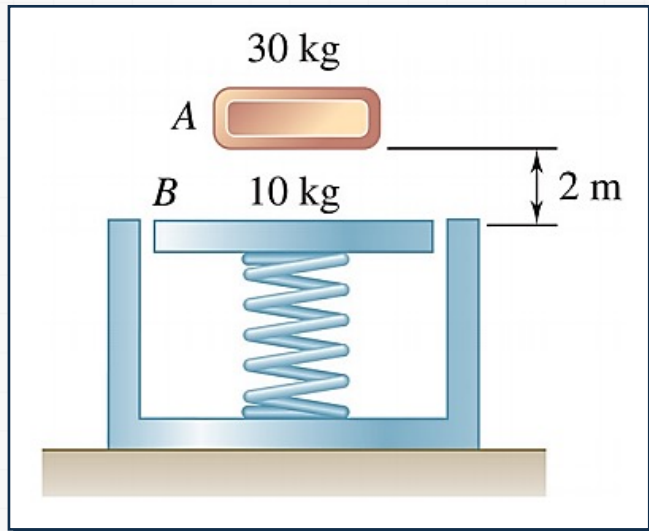


- Since $e = 1$, the impact between A and B is perfectly elastic.
- Therefore, rather than using the coefficient of restitution, you could have used the conservation of energy as your final equation.

7 Impulse and momentum principle

Sample Problem 5

⚙ Determine the maximum deflection of the pan



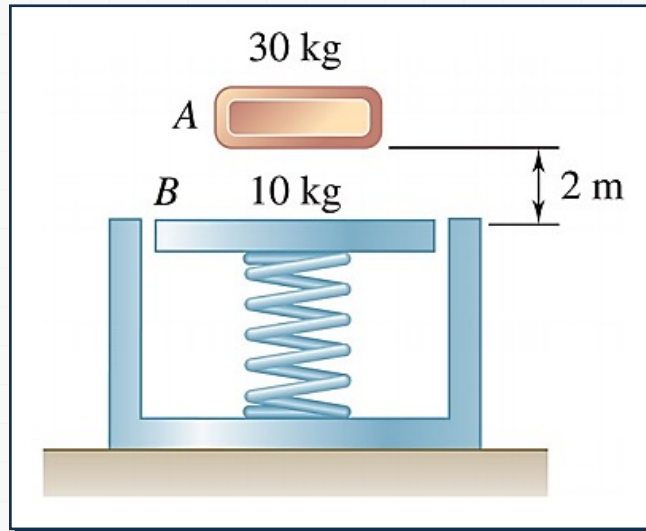
- A 30 kg block is dropped from a height of 2 m onto the 10 kg pan of a spring scale.
- Assuming the impact to be perfectly plastic, determine the maximum deflection of the pan. The constant of the spring is $k = 20\text{ kN/m}$.

7 Impulse and momentum principle

Sample Problem 5

⚙ Determine the maximum deflection of the pan

• Strategy



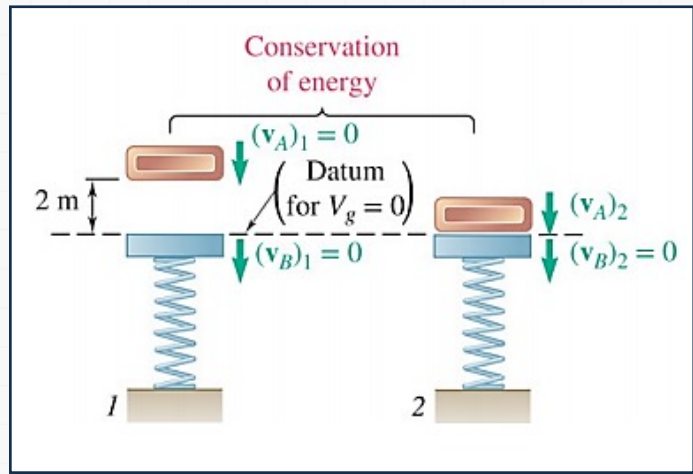
- Apply the principle of conservation of energy to determine the velocity of the block at the instant of impact.
- Since the impact is perfectly plastic, the block and pan move together at the same velocity after impact. Determine that velocity from the requirement that the total momentum of the block and pan is conserved.
- Apply the principle of conservation of energy to determine the maximum deflection of the spring.

7 Impulse and momentum principle

Sample Problem 5

⚙️ **Determine the maximum deflection of the pan**

• Modeling and Analysis



- Apply principle of conservation of energy to determine velocity of the block at instant of impact.

$$T_1 = 0 \quad V_1 = W_A y = (30)(9.81)(2) = 588 \text{ J}$$

$$T_2 = \frac{1}{2} m_A (v_A)_2^2 = \frac{1}{2} (30) (v_A)_2^2$$

$$V_2 = 0$$

$$T_1 + V_1 = T_2 + V_2$$

$$0 + 588 = \frac{1}{2} (30) (v_A)_2^2 + 0$$

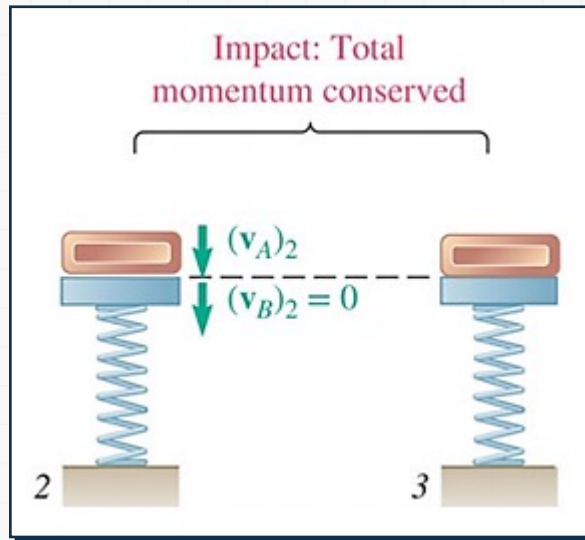
$$(v_A)_2 = 6.26 \text{ m/s}$$

7 Impulse and momentum principle

Sample Problem 5

⚙ Determine the maximum deflection of the pan

• Modeling and Analysis



- Determine velocity after impact from requirement that total momentum of the block and pan is conserved.

$$m_A(v_A)_2 + m_B(v_B)_2 = (m_A + m_B)v_3$$

$$(30)(6.26) + 0 = (30 + 10)v_3$$

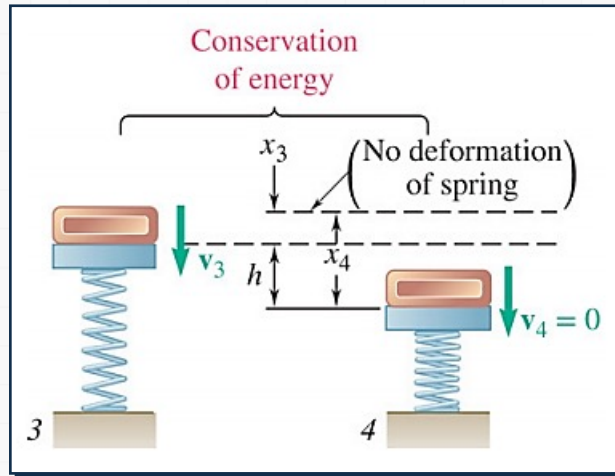
$$v_3 = 4.70\text{m/s}$$

7 Impulse and momentum principle

Sample Problem 5

⚙️ Determine the maximum deflection of the pan

• Modeling and Analysis



■ Initial spring deflection due to pan weight:

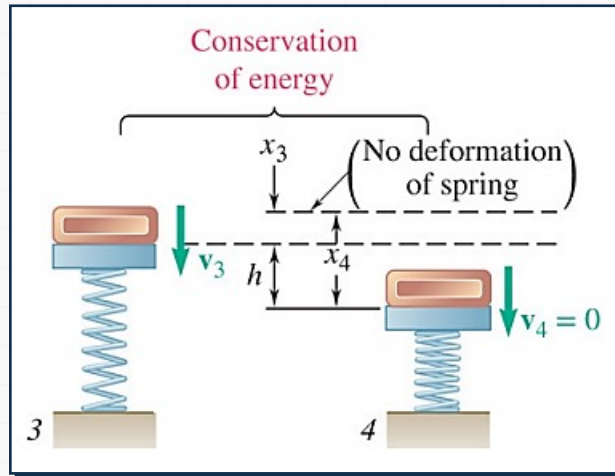
$$x_3 = \frac{W_B}{k} = \frac{(10)(9.81)}{20 \times 10^3} = 4.91 \times 10^{-3} \text{ m}$$

7 Impulse and momentum principle

Sample Problem 5

⚙ Determine the maximum deflection of the pan

• Modeling and Analysis



- Apply the principle of conservation of energy to determine the maximum deflection of the spring.

$$T_3 = \frac{1}{2}(m_A + m_B)v_3^2 = \frac{1}{2}(30 + 10)(4.7)^2 = 442 \text{ J}$$

$$V_3 = V_g + V_e$$

$$= 0 + \frac{1}{2}kx_3^2 = \frac{1}{2}(20 \times 10^3)(4.91 \times 10^{-3})^2 = 0.241 \text{ J}$$

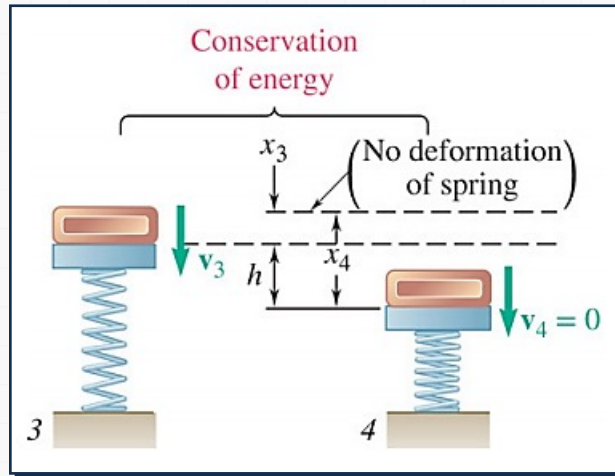
$$T_4 = 0$$

7 Impulse and momentum principle

Sample Problem 5

⚙ Determine the maximum deflection of the pan

• Modeling and Analysis



- Apply the principle of conservation of energy to determine the maximum deflection of the spring.

$$\begin{aligned}
 V_4 &= V_g + V_e = (W_A + W_B)(-h) + \frac{1}{2} k x_4^2 \\
 &= -392(x_4 - x_3) + \frac{1}{2}(20 \times 10^3)x_4^2 \\
 &= -392(x_4 - 4.91 \times 10^{-3}) + \frac{1}{2}(20 \times 10^3)x_4^2
 \end{aligned}$$

$$T_3 + V_3 = T_4 + V_4$$

$$442 + 0.241 = 0 - 392(x_4 - 4.91 \times 10^{-3}) + \frac{1}{2}(20 \times 10^3)x_4^2$$

$$x_4 = 0.230 \text{ m}$$

$$h = x_4 - x_3 = 0.230 - 4.91 \times 10^{-3}$$

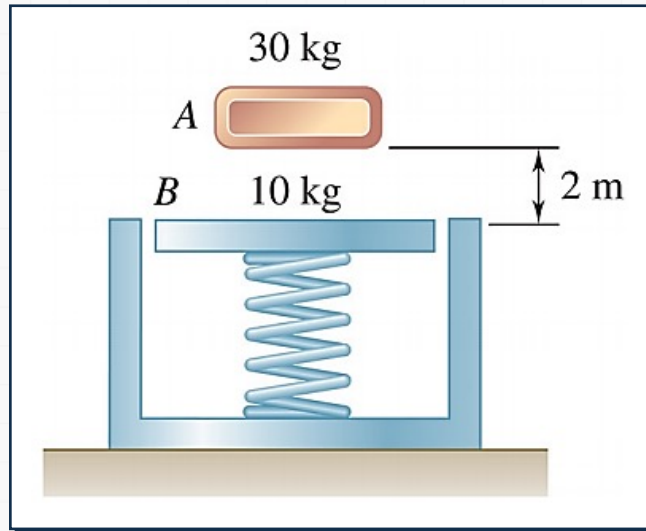
$$h = 0.225 \text{ m}$$

7 Impulse and momentum principle

Sample Problem 5

⚙️ **Determine the maximum deflection of the pan**

• Reflect and Think

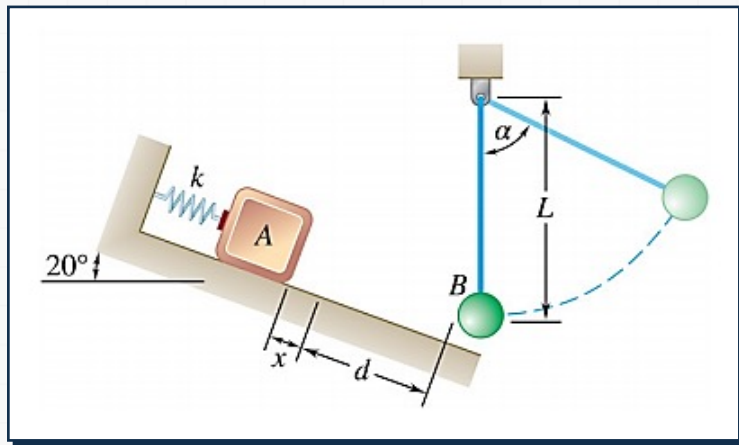


- The spring constant for this scale is pretty large, but the block is fairly massive and is dropped from a height of 2 m . From this perspective, the deflection seems reasonable.
- We included the spring in the system so we could treat it as an energy term rather than finding the work of the spring force.

7 Impulse and momentum principle

Group Problem Solving

⚙ Find the speed and tension

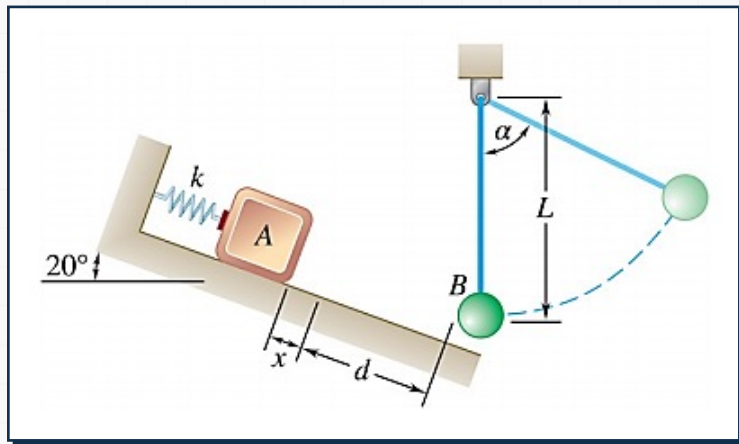


- A $2 - kg$ block A is pushed up against a spring compressing it a distance $x = 0.1m$.
- The block is then released from rest and slides down the 20° incline until it strikes a $1 - kg$ sphere B, which is suspended from a $1m$ inextensible rope.

7 Impulse and momentum principle

Group Problem Solving

⚙ Find the speed and tension



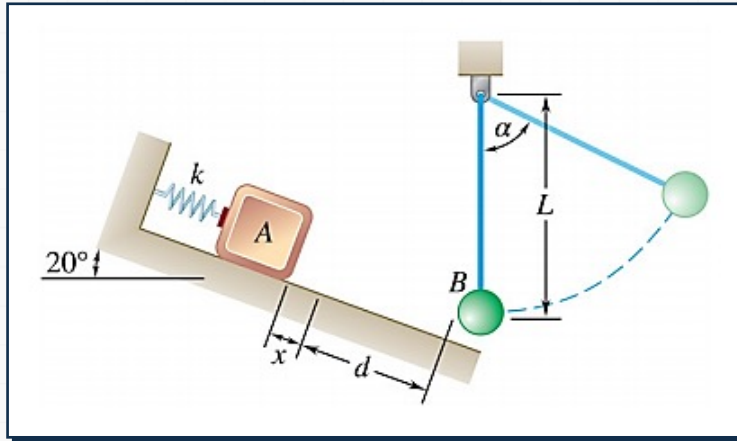
- The spring constant is $k = 800 \text{ N/m}$, the coefficient of friction between A and the ground is 0.2, the distance A slides from the unstretched length of the spring $d = 1.5 \text{ m}$, and the coefficient of restitution between A and B is 0.8. When $\alpha = 40^\circ$, find
 - (a) the speed of B
 - (b) the tension in the rope.

7 Impulse and momentum principle

Group Problem Solving

⚙ Find the speed and tension

• Strategy



- This is a multiple step problem. Formulate your overall approach.
- Use work-energy to find the velocity of the block just before impact.
- Use conservation of momentum to determine the speed of ball B after the impact.
- Use work energy to find the velocity at α .
- Use Newton's 2nd Law to find tension in the rope.

References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, Chapter 13